

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and infrastructure Engineering) SEMESTER - 5 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and infrastructure Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCIP506B - APPLIED GEOLOGY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 The oldest era of geological time scale is:
 - a) Mesozoic
 - b) Cenozoic
 - c) Paleozoic
 - d) Precambrian
- 2 Vindhyan rocks are mainly:
 - a) Igneous
 - b) Metamorphic
 - c) Sedimentary
 - d) Volcanic
- 3 Exploratory drilling is carried out to obtain information about:
 - a) Surface features
 - b) Sub-surface conditions
 - c) Climate
 - d) Vegetation
- 4 An aquifer which transmits water freely is called:
 - a) Aquiclude
 - b) Aquifuge
 - c) Aquifer
 - d) Aquitard
- 5 Columnar joints are formed due to
 - a) Weathering
 - b) Cooling contraction
 - c) Folding
 - d) Faulting
- 6 Red bole represents
 - a) Lava flow
 - b) Soil horizon
 - c) Dyke rock
 - d) Plutonic rock
- 7 Most suitable basalt for construction is
 - a) Weathered basalt
 - b) Compact basalt
 - c) Vesicular basalt
 - d) Red bole

- 8 Residual soil is formed by
 - a) Transportation
 - b) Weathering in place
 - c) Deposition
 - d) Erosion
- 9 Electrical resistivity depends on
 - a) Density
 - b) Porosity
 - c) Electrical properties
 - d) Magnetism
- 10 Soils derived from basalt are generally
 - a) Sandy
 - b) Clayey
 - c) Saline
 - d) Peaty
- 11 Plate tectonics explains
 - a) Volcanism only
 - b) Earthquakes only
 - c) Crustal movements
 - d) Weathering
- 12 Rock mechanics deals with
 - a) Soil behaviour
 - b) Rock behaviour
 - c) Concrete
 - d) Steel

Q2. Solve the following.

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| A) Describe major divisions of geological time scale and important events. | 6 |
| B) Explain lithology and structure of Deccan Trap basalts. | 6 |

Q3. Solve the following.

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| A) Explain the methods of sub-surface exploration used in civil engineering. | 6 |
| B) Explain aquifer theory with types of aquifers. | 6 |

Q4. Solve Any Two of the following.

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|--|---|
| A) Explain the origin, types and engineering significance of joints in Deccan Trap basalts. | 6 |
| B) Describe the geological setting and evolution of the Deccan Trap volcanism with reference to engineering geology. | 6 |
| C) Explain the tail channel erosion problem in the Deccan Trap region with suitable examples. | 6 |

Q5. Solve Any Two of the following.

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| A) Discuss the characteristics and engineering significance of river sand from Deccan Trap region. | 6 |
| B) Discuss the geological classification of soils with suitable examples. | 6 |
| C) Describe Wenner configuration with neat sketch and explain its field procedure. | 6 |

Q6. Solve Any Two of the following.

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|--|---|
| A) Explain the scope and importance of rock mechanics in civil engineering projects. | 6 |
| B) Discuss the geological and anthropogenic factors responsible for landslides. | 6 |
| C) Explain plate tectonics theory and its significance in understanding earthquakes. | 6 |

*** End ***